

Jasmine Wiggley

Washington University of St. Louis

Risa 3D

Analysis of a Stable 3D Model Structure

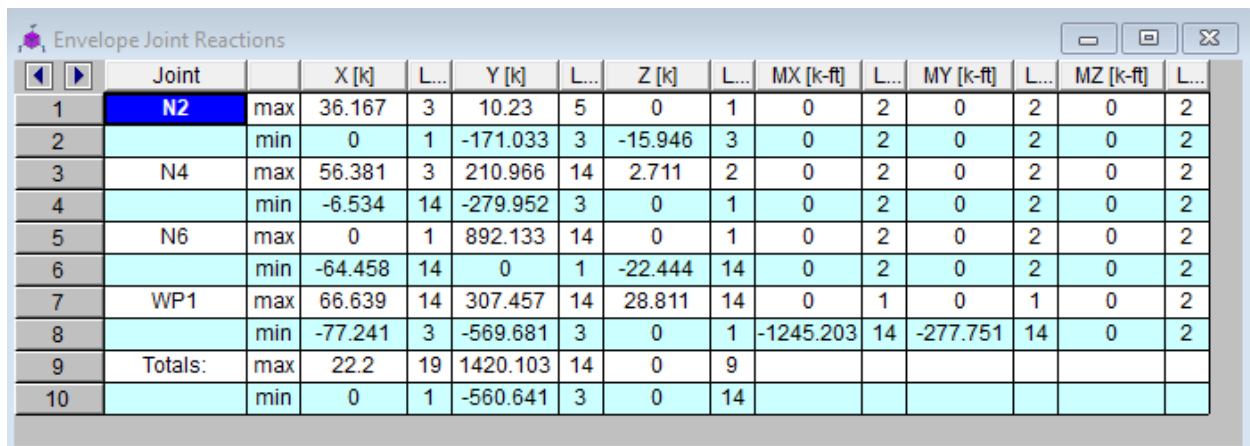
Analysis of a Stable 3D Model Structure

Procedure:

Create a Stable 3D Model Structure and compare Risa 3D results to handwritten calculations.

Results:

The following tables (joint coordinate reactions [Table 1](#)) were used in the creation of the 3D Model Structure in Figure 1.



	Joint		X [k]	L...	Y [k]	L...	Z [k]	L...	MX [k-ft]	L...	MY [k-ft]	L...	MZ [k-ft]	L...
1	N2	max	36.167	3	10.23	5	0	1	0	2	0	2	0	2
2		min	0	1	-171.033	3	-15.946	3	0	2	0	2	0	2
3	N4	max	56.381	3	210.966	14	2.711	2	0	2	0	2	0	2
4		min	-6.534	14	-279.952	3	0	1	0	2	0	2	0	2
5	N6	max	0	1	892.133	14	0	1	0	2	0	2	0	2
6		min	-64.458	14	0	1	-22.444	14	0	2	0	2	0	2
7	WP1	max	66.639	14	307.457	14	28.811	14	0	1	0	1	0	2
8		min	-77.241	3	-569.681	3	0	1	-1245.203	14	-277.751	14	0	2
9	Totals:	max	22.2	19	1420.103	14	0	9						
10		min	0	1	-560.641	3	0	14						

Table 1: Joint Reactions

